

1. (Section 6.4, Problem 2) Find a new representation of the quadratic equation $5x^2 + 6xy + 5y^2 - 8 = 0$ after rotating through the angle $\theta = 45^\circ$. Sketch the curve, showing both the old and new coordinate systems.

$$\begin{cases} x = \frac{x' - y'}{\sqrt{2}} \\ y = \frac{x' + y'}{\sqrt{2}} \end{cases}$$

$$5 \left(\frac{x' - y'}{\sqrt{2}} \right)^2 + 6 \left(\frac{x' - y'}{\sqrt{2}} \right) \left(\frac{x' + y'}{\sqrt{2}} \right) + 5 \left(\frac{x' + y'}{\sqrt{2}} \right)^2 = 8$$

$$\frac{5}{2} (x'^2 - 2x'y' + y'^2) + \frac{6}{2} (x'^2 + x'y' - x'y' - y'^2) + \frac{5}{2} (x'^2 + 2x'y' + y'^2) = 8$$

$$\left(\frac{5}{2} + \frac{6}{2} + \frac{5}{2} \right) x'^2 + \left(-5x'y' + 5x'y' \right) + \left(\frac{5}{2} - \frac{6}{2} + \frac{5}{2} \right) y'^2 = 8$$

$$8x'^2 + 2y'^2 = 8$$

$$\frac{x'^2}{1} + \frac{y'^2}{4} = 1$$

$$b^2 = 1 \quad a^2 = 4$$

$$b = 1 \quad a = 2$$

$$c^2 = a^2 - b^2 = 4 - 1 = 3$$

$$c = \sqrt{3}$$



